

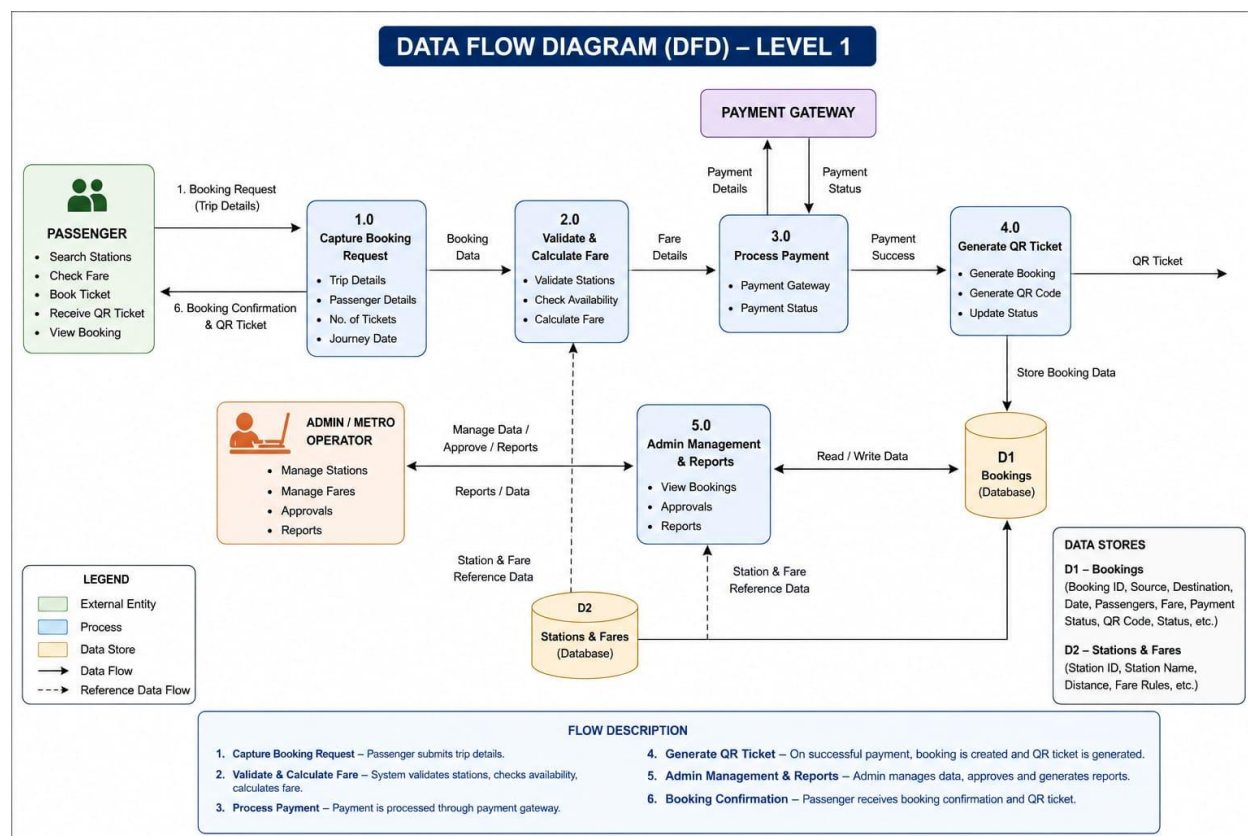
Project Design Phase-II

Data Flow Diagram & User Stories

Date	27 June 2026
Team ID	
Project Name	Metro Ticket Generation System Using ServiceNow
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. It shows how data enters and leaves the system, what changes the information, and where data is stored.



Level 0 DFD – Context Diagram:

External Entities: Commuter (inputs travel details, receives booking confirmation) and Administrator (manages and views booking records).

Central Process: Metro Ticket Generation System (ServiceNow Platform)

Data Flows: Commuter → System: Travel details (source station, destination, date, passengers). System → Commuter: Booking confirmation. Administrator → System: Access request for records. System → Administrator: Booking records and reports.

Level 1 DFD – Detailed Processes:

- Process 1 – User Authentication: Commuter login validated by ServiceNow user management.
- Process 2 – Metro Ticket Request Form: Captures source station, destination, travel date, number of passengers, and passenger details from the commuter.
- Process 3 – Input Validation: Client Scripts validate mandatory fields and flag errors in real-time.
- Process 4 – Fare Calculation Engine: Client Script calculates ticket fare based on selected route and passenger count.
- Process 5 – Flow Designer Automation: Triggered on form submission; captures catalog variables and initiates booking record creation.
- Process 6 – Metro Booking Table (Database): Stores completed booking records including all passenger and travel details.
- Process 7 – Admin Dashboard: Reads booking records from the Metro table and presents them to administrators.

User Stories:

User Type	Epic	USN	User Story	Acceptance Criteria	Priority	Release
Commuter	Metro Ticket Booking	USN-1	As a commuter, I can open the Metro Ticket Booking catalog item and fill in source station, destination, date, and passenger details.	Form loads successfully and all fields are visible	High	Sprint-1
		USN-2	As a commuter, I can see the fare automatically calculated when I select source and destination stations.	Fare displayed correctly before submission	High	Sprint-1
		USN-3	As a commuter, I receive a	Error message	High	Sprint-1

			validation error if I submit the form without filling mandatory fields.	shown; form not submitted		
		USN-4	As a commuter, I can submit the ticket booking form and receive a booking confirmation.	Booking record created in Metro table	High	Sprint-1
Administrator	Admin Dashboard	USN-5	As an admin, I can view all metro booking records in the ServiceNow list view with booking details.	All records visible with correct data	High	Sprint-1
		USN-6	As an admin, I can search and filter booking records by date, source, or destination.	Filter returns correct records	Medium	Sprint-2
System	Workflow Automation	USN-7	As the system, Flow Designer automatically creates a booking record when a commuter submits the ticket form.	Record created in Metro table without manual action	High	Sprint-1